# A Cheatsheet for Lie Algebra

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# Chapter 1

# Lie Algebra

## 1.1 Basic Concepts

**Definition 1.1.1.** *A vector space L over a field F, with an operation* $\begin{array}{ccc} L\times L & \to & L \\ (x,y) & \mapsto & [xy] \end{array}$ *called the bracket of* $x,y$ *is called a Lie algebra over F if the following axioms are satisfied :* $\begin{cases} \text{the bracket operation is bilinear} \\ [xx]=0,\ \forall x\in L \\ [x[yz]]+[y[zx]]+[z[xy]]=0,\ \forall x,y,z\in L \end{cases}$.

$\to$ $\begin{array}{rl} [x+y,x+y]=[xx]+[xy]+[yx]+[yy] & \text{by the axiom 1} \\ =0+[xy]+yx]+0 & \text{by the axiom 2} \end{array}$ and therefore $[xy]=-[yx]$ and the converse holds given char$F\neq 2$.

**Definition 1.1.2.** *Two Lie algebras* $L$,$L'$ *over* $F$ *are isomorphic if there exists a vector space isomorphism* $\phi: L\to L'$ *satisfying* $\phi([xy])=[\phi(x)\phi(y)],\ \forall x,y\in L$.

**Definition 1.1.3.** *A subspace K of L is called a subalgebra if* $[xy]\in K,\ \forall x,y\in K$.

$\to \forall 0\neq x\in L$ it defines a 1-dimensional subalgebra $Fx$ with trivial multiplication by the axiom 2 of the definition.

If $V$ is a finite dimensional vector space over $F$, denoted by End$V$, the set of linear transformations $V\to V$. As a vector space over $F$, End$V$ has dimension $n^2$ where $n=\dim V$, and End$V$ is a ring relative to the usual product operation. Define a new operation $[xy]=xy-yx$, called the bracket of $x,y$. With this operation, End$V$ is a Lie algebra over $F$.
pf. Axiom 1, 2 are immediate. $[x[yz]]+[y[zx]]+[z[xy]]=[x(yz-zy)]+[y(zx-xz)]+[z(xy-yx)]=[x(yz)]-[x(zy)]+[y(zx)]-[y(xz)]+[z(xy)]-[z(yx)]=x(yz)-(yz)x-x(zy)+(zy)x+y(zx)-(zx)y-y(xz)+(xz)y+z(xy)-(xy)z-z(yx)+(yx)z=0$ by the

associativity of multiplication operation. Hence the axiom 3 follows and therefore it is a Lie algebra.

Write $\mathfrak{gl}(V)$ for End$V$ viewed as Lie algebra and call it **the general linear algebra**. Any subalgebra of a Lie algebra $\mathfrak{gl}(V)$ is called a **linear Lie algebra.** Here $\mathfrak{gl}(V)$ can be identified with the set of all $n \times n$ matrices over $F$, denoted $\mathfrak{gl}(n,F)$.

e.g.

- Let $\dim V = l+1$. Denote by $\mathfrak{sl}(V) = \mathfrak{sl}(l+1,F)$ the set of endomorphisms of $V$ having trace zero. Since $\begin{cases} T(xy) = T(yx) \\ T(x+y) = T(x)+T(y) \end{cases}$, $\mathfrak{sl}(V)$ is a subalgebra of $\mathfrak{gl}(V)$, called the **special linear algebra**.

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  - when $\dim V = 2l$ with basis $(v_1,\ldots,v_{2l})$, denote by $\mathfrak{sp}(V) = \mathfrak{sp}(2l,F)$, the symplectic algebra, which by definition consists of all endomorphisms $x$ of $V$ satisfying $f(x(v),w) = -f(v,x(w))$.
  - when $\dim V = 2l+1$ with basis $(v_1,\ldots,v_{2l+1})$, denote by $\mathfrak{o}(V) = \mathfrak{o}(2l+1,F)$, the orthogonal algebra, which by definition consists of all endomorphisms $x$ of $V$ satisfying $f(x(v),w) = -f(v,x(w))$.

- $\begin{cases} \mathfrak{t}(n,F) \text{ the set of upper triangular matrices } (a_{ij}),\ a_{ij}=0,\ \forall i>j \\ \mathfrak{n}(n,F) \text{ the set of the strictly upper triangular matrices } (a_{ij}),\ a_{ij}=0,\ \forall i \geq j \\ \mathfrak{d}(n,F) \text{ the set of all diagonal matrices} \end{cases} \rightarrow$
  $\mathfrak{t}(n,F) = \mathfrak{d}(n,F) + \mathfrak{n}(n,F)$ with $[\mathfrak{d}(n,F)\mathfrak{n}(n,F)] = \mathfrak{n}(n,F)$.

- Lie algebra of derivation

  By an $F$-algebra, it is a vector space $\mathfrak{U}$ over $F$ endowed with a bilinear operation $\mathfrak{U} \times \mathfrak{U} \to \mathfrak{U}$. By a derivation of $\mathfrak{U}$, it is a linear map $\delta : \mathfrak{U} \to \mathfrak{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$. Then the collection Der$\mathfrak{U}$ of all derivation of $\mathfrak{U}$ is a vector subspace of End$\mathfrak{U}$. Also the bracket of two derivations $[\delta\delta']$ is again derivation, and therefore Der$\mathfrak{U}$ is a subalgebra of $\mathfrak{gl}(\mathfrak{U})$. Then since a Lie algebra $L$ is an $F$-algebra in the above sense, Der$L$ can be defined.
  e.g. If $x \in L$, $y \mapsto [xy]$ is an endomorphism of $L$, denoted ad$x$. Then ad$x \in$ Der$L$ by the Jacobi identity in the form $[x[yz]] = [[xy]z] + [y[xz]]$. Derivation of this form are called **inner**, all others **outer**. The map $\begin{array}{ccc} L & \to & \mathrm{Der}L \\ x & \mapsto & \mathrm{ad}x \end{array}$ is called the **adjoint representation** of $L$.

If $L$ is an arbitrary finite dimensional vector space over $F$, we can view $L$ as a Lie algebra by setting $[xy] = 0,\ \forall x, y \in L$. Such an algebra is called **abelian**. If $L$ is any Lie algebra, with basis $x_1, \ldots, x_n$, the entire multiplication table of $L$ can be recovered from the structure constants $a_{ij}^k$ s.t. $[x_i x_j] = \sum_{k=1}^n a_{ij}^k x_k$. Then by the axiom 2,3 of the definition gives $\left\{ \begin{array}{c} a_{ii}^k = 0 = a_{ij}^k + a_{ji}^k \\ \sum_k (a_{ij}^k a_{kl}^m + a_{jl}^k a_{ki}^m + a_{li}^k a_{kj}^m) = 0 \end{array} \right.$.

A subspace $I$ of a Lie algebra $L$ is called an **ideal** of $L$ if $\forall x \in L,\ \forall y \in I,\ [xy] \in I$. Then $0, L$ are ideals of $L$ trivially. The centre $Z(L) = \{z \in L | [xz] = 0,\ \forall x \in L\}$ is also an ideal of $L$. If $I, J$ are two ideals of a Lie algebra $L$, then $\left\{ \begin{array}{c} I + J = \{x + y | x \in I, y \in J\} \\ [IJ] = \{\sum x_i y_i | x_i \in I, y_i \in J\} \end{array} \right.$ are ideals.

**Definition 1.1.4.** *If $L$ has no ideals except itself and 0, and moreover $[LL] \neq 0$, $L$ is caled simple.*

$\rightarrow L$ is simple$\rightarrow Z(L) = 0,\ L = [LL]$.

e.g. $L = \mathfrak{sl}(2, F),\ \mathrm{char}F \neq 2$. Take the standard basis of $L$, $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $\left\{ \begin{array}{c} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{array} \right.$. If $I \neq 0$ is an ideal of $L$, let $ax + by + ch$ be an arbitrary nonzero element of $I$. Applying ad$x$ twice, we get $-2bx \in I$, and applying ad$y$ twice, we get $-2ay \in I$. Thus if $a, b \neq 0$, $y \in I$ or $x \in I$ and clearly $I = L$. If $a = b = 0$, $0 \neq ch \in I \rightarrow h \in I$ and $I = L$. So $L$ is simple.

The construction of a quotient algebra $L/I$ is same as the construction of the quotient ring. As vector space $L/I$ is the quotient space, but Lie quotient algebra $L/I$ has an extra structure $[x + I, y + I] = [xy] + I$.

The normaliser of a subalgebra $K$ of $L$ is defined by $N_L(K) = \{x \in L | [xK] \subset K\}$. Then by the Jacobi identity, it is a subalgebra of $L$. If $K = N_L(K)$, we call $K$ self-normalising.

**Definition 1.1.5.** *Given two Lie algebras $L, L'$, a linear transformation $\phi : L \rightarrow L'$ is called a homomorphism if $\phi([xy]) = [\phi(x)\phi(y)],\ \forall x, y \in L$. $\phi$ is called a monomorphism if $\ker\phi = 0$, an epimorphism if $Im\phi = L'$, an isomorphism if it is both monomorphism and epimorphism.*

$\rightarrow \left\{ \begin{array}{r} \ker\phi \text{ is an ideal of } L (\forall x \in \ker\phi, \forall y \in L, \phi([xy]) = [\phi(x)\phi(y)] = 0) \\ \mathrm{Im}\phi \text{ is a subalgebra of } L' \end{array} \right.$.

**Proposition 1.1.1.** *1. If $\phi : L \to L'$ is a homomorphism of Lie algebras, then $L/\ker\phi \cong Im\phi$. If $I$ is any ideal of $L$ included in $\ker\phi$, there exists a unique homomorphism $\psi : L/I \to L'$ making the diagram*

$$\begin{tikzcd} L \arrow[r, "\phi"] \arrow[dr, "\pi"'] & L' \\ & L/I \arrow[u, "\psi"'] \end{tikzcd}$$

*commutes.*

2. *If $I, J$ are ideals of $L$ s.t. $I \subset J$, $J/I$ is an ideal of $L/I$ and $(L/I)/(J/I)$ is naturally isomorphic to $L/J$.*

3. *If $I, J$ are ideals of $L$, there is a natural isomorphism between $(I+J)/J$ and $I/(I \cap J)$.*

pf. by the general results of linear algebras.

**Definition 1.1.6.** *A representation of a Lie algebra $L$ is a homomorphism $\phi : L \to \mathfrak{gl}(V)$.*

e.g. the adjoint representation $\begin{array}{rcc} \mathrm{ad} : L & \to & \mathfrak{gl}(L) \\ x & \mapsto & \mathrm{ad}x \end{array}$. Then $[\mathrm{ad}x\mathrm{ad}y](z) = \mathrm{ad}x\mathrm{ad}y(z) - \mathrm{ad}y\mathrm{ad}x(z) = \mathrm{ad}x([yz]) - \mathrm{ad}y([xz]) = [x[yz]] - [y[xz]] = [x[yz]] + [[xz]y] = [[xy]z] = \mathrm{ad}[xy](z)$ so it preserves the bracket. The kernel of ad consists of $x \in L$ for which $\mathrm{ad}x = 0$, i.e. for which $[xy] = 0,\ \forall y \in L$. So $\ker\mathrm{ad} = Z(L)$.

**Definition 1.1.7.** *An automorphism of $L$ is an isomorphism of $L$ onto itself. AutL denotes the group of all such.*

e.g. If $g \in \mathrm{GL}(V)$ is any invertible endomorphism of $V$ and $gLg^{-1} = L$, then the map $x \mapsto gxg^{-1}$ is an automorphism of $L$.

Assume char$F = 0$. Suppose $x \in L$ is an element for which ad$x$ is nilpotent, i.e. $(\mathrm{ad}x)^k = 0$, $k > 0$. Then the exponential power series for a linear transformation over $\mathbb{C}$ makes sense over $F$, because it has only finitely many terms; $\exp(\mathrm{ad}x) = 1 + \mathrm{ad}x + \frac{(\mathrm{ad}x)^2}{2!} + \cdots + \frac{(\mathrm{ad})^{k-1}}{(k-1)!}$.

**CLAIM : If ad$x$ is replaced by an arbitrary nilpotent derivation $\delta$ of $L$, then $\exp(\delta) \in$ Aut$L$.**

pf.

$$\begin{aligned} \exp(\mathrm{ad}x)\exp(\mathrm{ad}y) &= \exp(\delta(x))\exp(\delta(y)) && \text{by hypothesis} \\ &= \left(\sum_{i=0}^{k-1}\left(\frac{\delta^i x}{i!}\right)\right)\left(\sum_{j=0}^{k-1}\left(\frac{\delta^j y}{j!}\right)\right) && = \sum_{n=0}^{2k-2}\left(\sum_{i=0}^{n}\left(\frac{\delta^i x}{i!}\right)\left(\frac{\delta^{n-i} y}{(n-i)!}\right)\right) \\ &= \sum_{n=0}^{2k-2} \frac{\delta^n(xy)}{n!} && \text{by Leibniz rule} \\ &= \sum_{n=0}^{k-1} \frac{\delta^n(xy)}{n!} && \text{since } \delta^k = 0 \\ &= \exp(\delta(xy)) \end{aligned}$$

. Then $\exp\delta$ is invertible since its inverse is $1 - \eta + \eta^2 = \cdots \pm \eta^{k-1}$, $\exp\delta = 1 + \eta$.

An automorphism of the form $\exp(\mathrm{ad}x)$ where $\mathrm{ad}x$ is nilpotent is called **inner**. The subgroup of $\mathrm{Aut}L$ generated by these inner automorphisms is denoted $\mathrm{Int}L$ and it s a normal subgroup ; $\forall \phi \in \mathrm{Aut}L,\ \forall x \in L,\ \phi(\mathrm{ad}x)\phi^{-1} = \mathrm{ad}\phi(x)$ and therefore $\phi(\exp\mathrm{ad}x)\phi^{-1} = \exp(\mathrm{ad}\phi(x))$.
e.g. Let $L = \mathfrak{sl}(2,F)$ with standard basis $(x,y,h)$. Define $\sigma = \exp\mathrm{ad}x\cdot\exp\mathrm{ad}(-y)\exp\mathrm{ad}x \in$ $\mathrm{Int}L$. Then $\begin{cases} \sigma(x) = -y \\ \sigma(y) = -x \\ \sigma(h) = -h \end{cases}$ with the order of $\sigma$ 2.
$\rightarrow$ if $L \subset \mathfrak{gl}(V)$ is an arbitrary linear Lie algebra with char $F = 0,\ x \in L$ is nilpotent, then

$$(\exp x)y(\exp x)^{-1} = \exp\mathrm{ad}x(y),\quad \forall y \in L$$

pf. $\mathrm{ad}x = \lambda_x + \rho_{-x}$ where $\lambda_x, \rho_x$ denote the left and right multiplication by $x$ in the ring $\mathrm{End}V$. Then $\exp\mathrm{ad}x = \exp(\lambda_x + \rho_{-x}) = \exp\lambda_x \cdot \exp\rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp(-x)}$ so the equation follows.

**Definition 1.1.8.** *Define a sequence of ideals of L called the derived series by* $L^{(0)} = L,\ L^{(1)} = [LL],\ L^{(2)} = [L^{(1)}L^{(1)}],\ldots,L^{(i)} = [L^{(i-1)}L^{(i-1)}]$. *Call L solvable if* $L^{(n)} = 0$ *for some n.*

e.g. $L = \mathfrak{t}(n,f)$ **is solvable.**
pf. The basis for $\mathfrak{t}(n,F)$ consists of the matrix units $e_{ij},\ i \le j$ and the dimension is $\frac{n(n+1)}{2}$. Then $[e_{ii}, e_{il}] = e_{il},\ i < l$, thus $\mathfrak{n}(n,f) \subset [LL]$, where $\mathfrak{n}(n,F)$ is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n,F) = \mathfrak{d}(n,F) + \mathfrak{n}(n,F)$, with $\mathfrak{d}(n,F)$ abelian, $\mathfrak{n}(n,F)$ is equal to the derived algebra of $L$. In $\mathfrak{n}(n,F)$, we have a natural notion of level for $e_{ij}$, namely $j - i$. Assume $i < j, k < l$. Wlog assume $i \ne l$. Then $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. Each $e_{il}$ is commutator of two matrices whose levels add upto that of $e_{il}$. thus $L^{(2)}$ is spanned by those $e_{ij}$ of level $\ge 2$, $L^{(i)}$ by those of level $\ge 2^{i-1}$. Then it is clear that $L^{(i)} = 0$ whenever $2^{i-1} > n - 1$. The claim follows.

**Proposition 1.1.2.** *Let L be a Lie algebra.*

1. *If L is solvable, then so are all subalgebras and homomorphic images of L.*
2. *If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable.*
3. *IfI, J are solvable ideals of L, then so is* $I + J$.

pf.

- By the definition, if $K$ is a subalgebra of $L$, $K^{(i)} \subset L^{(i)}$. Similarly, if $\phi : L \rightarrow M$ is an epimorphism, $\phi(L^{(i)}) = M^{(i)}$ gives the stmt.

- Say $(L/I)^{(n)} = 0$. Then by 1 the canonical homomorphism $\pi : L \to L/I$ has $\pi(L^{(n)}) = 0$ i.e. $L^{(n)} \subset I = \ker \pi$. If $I^{(m)} = 0$, then $(L^{(i)})^{(j)} = L^{(i+j)}$ implies $L^{(n+m)} = 0$.

- An isomorphism between $(I+J)/J$ and $I/(I \cap J)$ exists. As a homomorphic image of $I$, $I/(I \cap J)$ is solvable, and therefore its isomorphic image $(I+J)/J$ is solvable. Then so is $I+J$ by 2.

$\to$ let $S$ be a maximal solvable ideal of the Lie algebra $L$. If $I$ is any solvable ideal of $L$, by 3, $S+I = S$ by maximality or $I \subset S$. Thus there exists the unique maximal solvable ideal, called **the radical** of $L$, denoted Rad$L$. If Rad$L = 0$, $L$ is called **semisimple.**.

**Definition 1.1.9.** *Define a sequence of ideals of L called the descending central series by* $L^0 = L, L^1 = [LL], L^2 = [LL^1], \ldots, L^i = [LL^{i-1}]$. *L is called nilpotent if* $L^n = 0$ *for some n.*

$\to L^{(i)} \subset L^i$, $\forall i$, and therefore nilpotent algebras are solvable, but not vice versa.

**Proposition 1.1.3.** *Let L be a Lie algebra.*

1. *If L is nilpotent, then so are all subalgebras and homomorphic images of L*

2. *If* $L/Z(L)$ *is nilpotent, then so is L*

3. *If L is nilpotent and nonzero,* $Z(L) \neq 0$.

pf.

- By the definition, for any subalgebra $K$ of $L$, $K^i \subset L^i$. Similarly, if $\phi : L \to M$ is an epimorphism, $\phi(L^i) = M^i$ gives the stmt.

- If $L^n \subset Z(L)$, $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$.

- The last nonzero term of the descending central series is central.

$\to \exists n$ s.t. $\mathrm{ad}x_1, \ldots, \mathrm{ad}x_n(y) = 0$, $\forall x_i, y \in L$. If $L$ is any Lie algebra and $x \in L$, we call $x$ **ad-nilpotent** if ad$x$ is a nilpotent endomorphism. If $L$ is nilpotent, then all elements of $L$ are ad-nilpotent.

**Lemma 1.1.1.** *Let* $x \in \mathfrak{gl}(V)$ *be a nilpotent endomorphism. Then adx is also nilpotent.*

pf. We may associate to $x$ two endomorphisms of End$V$; the left and right translation $\begin{cases} \lambda_x(y) = xy \\ \rho_x(y) = yx \end{cases}$, which are nilpotent because $x$ is. Moreover, $\lambda_x, \rho_x$ commute. In any ring including End$(V)$, the sum or difference of two commuting nilpotents is again nilpotent, so ad$x = \lambda_x - \rho_x$ is nilpotent.

**Theorem 1.1.1.** *Let $L$ be a subalgebra of $\mathfrak{gl}(V)$, $V$ finite dimensional. If $L$ consists of nilpotent endomorphisms and $V \neq 0$, $\exists 0 \neq v \in V$ for which $Lv = 0$.*

pf. When $\dim L = 0$ or 1, the theorem holds trivially.
When $\dim L > 1$, suppose $K \neq L$ be any subalgebra of $L$. Then by the previous lemma, $K$ acts as a Lie algebra of nilpotent linear transformations on the vector space $L$, hence also on the vector space $L/K$. Because $\dim K < \dim L$, by the inductive hypothesis, there exists a vector $x \in L/K$ s.t. $x + K \neq K$ s.t. it is killed by the image of $K$ in $\mathfrak{gl}(L/K)$. This means $[yx] \in K$, $\forall y \in K, x \notin K$, i.e. $K$ properly included in $N_L(K)$. Take $K$ to be a maximal proper subalgebra of $L$. Then $N_L(K) = L$, i.e. $K$ is an ideal of $L$. If $\dim L/K$ were greater than 1, then the inverse image in $L$ of a 1-dimensional subalgebra of $L/K$ would be a proper subalgebra properly containing $K$, a contradiction. Thus $K$ has codimension 1. Thus we can write $L = K + Fz$, where $z \in L \setminus K$. By induction, $W = \{v \in V | Kv = 0\}$ is nonzero. Since $K$ is an ideal, $W$ is stable under $L$ ( $\begin{cases} \forall x \in L, \\ \forall y \in K \end{cases}$, if $w \in W$, $yxw = xyw - [xy]w = 0$). Choose $z \in L \setminus K$ as above, so the nilpotent endomorphism $z$ has an eigenvector, i.e. $\exists 0 \neq v \in W$ for which $zv = 0$. Then $Lv = 0$ as desired.

**Theorem 1.1.2** (Engel)**.** *If all elements of $L$ are ad-nilpotent, then $L$ is nilpotent.*

pf. Given a Lie algebra $L$, all of whose elements are ad-nilpotent, ad$L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus there exists $x \neq 0$ in $L$ for which $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ evidently consists of ad-nilpotent elements and has smaller dimension than $L$. By the inductive hypothesis, $L/Z(L)$ is nilpotent. By the previous proposition 2, $L$ is nilpotent as well.

**Definition 1.1.10.** *If $V$ is a finite dimensional vector space, a flag in $V$ is a chain of subspaces $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$, $\dim V_i = i$.*

$\to$ if $x \in \mathrm{End}V$, $x$ is said to stabilises this flag provided $xV_i \subset V_i$, $\forall i$.

**Corollary 1.1.1.** *Under the hypotheses of the theorem, there exists a flag $(V_i)$ in $V$ stable under $L$, with $xV_i \subset V_{i-1}$, $\forall i$. In other words, there exists a basis of $V$ relative to which the matrices of $L$ are all in $\mathfrak{n}(n,F)$.*

pf. Begin with any nonzero $v \in V$ killed by $L$, the existence of which is assured by the theorem. Set $V_1 = Fv$. Let $W = V/V_1$ and the induced action of $L$ on $W$ is also by nilpotent endomorphisms. By the inductive hypotheses, $W$ has a flag stabilised by $L$, whose inverse image in $V$ does the trick.

**Lemma 1.1.2.** *Let $L$ be nilpotent, $K$ be an ideal of $L$. Then if $K \neq 0$, $K \cap Z(L) \neq 0$.*

pf. $L$ acts on $K$ via the adjoint representation, so by the previous theorem, $\exists 0 \neq x \in K$ killed by $L$, i.e. $[Lx] = 0 \to x \in K \cap Z(L)$.

## 1.2 Semisimple Lie Algebras

**Theorem 1.2.1.** *Let $L$ be a solvable subalgebra of $\mathfrak{gl}(V)$, $V$ finite dimensional. If $V \neq 0$, then $V$ contains a common eigenvector for all the endomorphisms in $L$.*

pf. When $\dim L = 0$, the theorem trivially holds. When $\dim L > 0$, $L$ is solvable of positive dimension, so $L$ properly includes $[LL]$. Since $L/[LL]$ is abelian, any subspace is an ideal. Take a subspace of codimension 1, then its inverse image $K$ is an ideal of codimension 1 in $L$. Then by the inductive hypothesis $\exists v \in V$ for $K$ a common eigenvector, i.e. $\forall x \in K,\ xv = \lambda(x)v$. Fix this linear function $\lambda : K \to F$, and denote by $W$ the subspace $\{w \in V | xw = \lambda(x)w,\ \forall x \in K\}$. Then $W \neq 0$.

- **$L$ leaves $W$ invariant**

  pf. Fix $w \in W, x \in L$. Let $n$ be the smallest positive integer s.t. $w, xw, \ldots, x^n w$ are L.D. Let $W_i$ be the subspace of $V$ spanned by $w, xw, \ldots, x^{i-1}w$ so that $\dim W_n = n,\ W_n = W_{n+i},\ \forall i \geq 0$ and $x$ maps $W_n$ into $W_n$. Then $\forall y \in K$ leaves each $W_i$ invariant. Relative to the basis $w, xw, \ldots, x^{n-1}w$ of $W_n$, $yx^i w = yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w = x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w$ where $w' \in W_{i-1}$ by the inductive hypothesis and since $x$ maps $W_{i-1}$ into $W_i$, $yx^i w \equiv \lambda(y)x^i w \mod W_i$. Then $T_{W_n}(y) = n\lambda(y)$. In particular, it is true for elements of $K$ of the special form $[xy]$. But $x, y$ both stabilise $W_n$, so $[xy]$ acts on $W_n$ as the commutator of two endomorphisms of $W_n$ and therefore its trace is 0. So $n\lambda([xy]) = 0$. Since $\mathrm{char} F = 0$, $\lambda([xy]) = 0$ and $L$ stabilises $W$.

Then write $L = K + Fz$. Since $F$ is algebraically closed, we can find an eigenvector $v_0 \in W$ of $z$ for some eigenvalue. Then $v_0$ is a common eigenvector for $L$ and $\lambda$ can be extended to a linear function on $L$ s.t. $xv_0 = \lambda(x)v_0,\ x \in L$.

**Corollary 1.2.1** (Lie's Theorem)**.** *Let $L$ be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then $L$ stabilises some flag in $V$, i.e. the matrices of $L$ relative to a suitable basis of $V$ are upper triangular.*

pf. by the induction on $\dim V$ in the use of the previous theorem.
$\to$ let $L$ be any solvable Lie algebra, $\phi : L \to \mathfrak{gl}(V)$ a finite dimensional representation of $L$. Then $\phi(L)$ is solvable by the previous proposition, hence stabilises a flag by the previous corollary.

**Corollary 1.2.2.** *Let $L$ be solvable. Then there exists a chain of ideals of $L$, $0 = L_0 \subset \cdots \subset L_n = L$ s.t. $\dim L_i = i$.*

**Corollary 1.2.3.** *Let $L$ be solvable. Then $x \in [LL]$ implies $ad_L x$ is nilpotent. In particular, $[LL]$ is nilpotent.*

pf. find a flag of ideals by the previous corollary. Relative to a basis $(x_1,\ldots,x_n)$ of $L$ for which $(x_1,\ldots,x_i)$ spans $L_i$, the matrices of ad$L$ lie in $\mathfrak{t}(n,F)$. Therefore the matrices of $[\mathrm{ad}L\,\mathrm{ad}L] = \mathrm{ad}_L[LL]$ lie in $\mathfrak{n}(n,F)$, the derived algebra of $\mathfrak{t}(n,F)$. Thus $\mathrm{ad}_L x$ is nilpotent for $x \in [LL]$. Then $\mathrm{ad}_{[LL]}x$ is nilpotent and so is $[LL]$ by Engel's theorem.

Call $x \in \mathrm{End}V$ **semisimple** if the roots of its minimal polynomial over $F$ are all distinct. Equivalently, $x$ is semisimple iff $x$ is diagonalisable, given $F$ being algebraically closed. If $x$ is semisimple and maps a subspace $W$ of $V$ into itself, then the restriction of $x$ to $W$ is semisimple.

**Proposition 1.2.1.** *Let $V$ be a finite dimensional vector space over $F$, $x \in EndV$.*

1. *There exists unique $x_s, x_n \in EndV$ satisfying the conditions : $x = x_s + x_n$, $x_s$ is semisimple, $x_n$ is nilpotent and $x_s, x_n$ commute.*

2. *There exits polynomials $p(T), q(T)$ in one indeterminate, without constant term, s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, $x_s, x_n$ commute with any endomorphism commuting with $x$.*

3. *If $A \subset B \subset V$ are subspaces and $x$ maps $B$ into $A$, then $x_s, x_n$ also maps $B$ into $A$.*

pf.

- 2

  Let $a_1,\ldots,a_k$ be the distinct eigenvalues of $x$ with multiplicities $m_1,\ldots,m_k$. Then the characteristic polynomial is $\Pi(T-a_i)^{m_i}$. If $V_i = \ker(x-a_i 1)^{m_i}$, then $V$ is the direct sum of the subspaces $V_1,\ldots,V_k$, each stable under $x$. On $V_i$, $x$ clearly has characteristic polynomial $(T-a_i)^{m_i}$. Then by the CRT, we have a polynomial $p(T) \equiv \begin{cases} a_i & \mod (T-a_i)^{m_i} \\ 0 & \mod T \end{cases}$. Set $q(T) = T - p(T)$. Then both $p(T), q(T)$ have no constant term. Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in $x$, $x_s, x_n$ commute with each other as well as all endomorphisms which commute with $x$.

- 3

  $x_s, x_n$ stabilise all subspaces of $V$ stabilised by $x$, in particular the $V_i$. The congruence $p(T) \equiv a_i \mod (T-a_i)^{m_i}$ shows the restriction of $x_s - a_i 1$ to $V_i$ is zero for all $i$, hence $x_s$ acts diagonally on $V_i$ with single eigenvalue $a_i$. By definition, $x_n = x - x_s$, so $x_n$ is nilpotent trivially. Since $p(T), q(T)$ have no constant term, 3 follows.

- 1

  Let $x = s + n$ be another such decomposition, so $x_s - s = n - x_n$. By 2, all endomorphisms commute. Sums of commuting semisimple and nilpotent endomorphisms are semisimple and nilpotent respectively. Thus $s = x_s, n = x_n$ since only 0 can be both semisimple and nilpotent.

→ the decomposition $x = x_s + x_n$ is called **the additive Jordan-Chevalley decomposition of** $x$. $x_s, x_n$ are called the semisimple part and the nilpotent part of $x$ respectively.

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, $V$ finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is $\mathrm{ad}x$ by the previous lemma.
**CLAIM : If $x$ is semisimple, so is ad$x$.**
pf. Choose a basis $(v_1, \ldots, v_n)$ of $V$ relative to which $x$ has matrix diag$(a_1, \ldots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \ldots, v_n)$ s.t. $e_{ij}(v_k) = \delta_{jk} v_i$. Then $\mathrm{ad}x(e_{ij}) = (a_i - a_j)e_{ij}$, so ad$x$ has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

**Lemma 1.2.1.** *Let $x \in EndV$, $x = x_s + x_n$ be its Jordan decomposition. Then $adx = adx_s + adx_n$ is the Jordan decomposition of $adx$.*

pf. $\mathrm{ad}x_s$ is semisimple and $\mathrm{ad}x_n$ is nilpotent, and they commute, since $[\mathrm{ad}x_s \mathrm{ad}x_n] = \mathrm{ad}[x_s x_n] = 0$. Then by the previous proposition 1, the lemma follows.

**Lemma 1.2.2.** *Let $\mathfrak{U}$ be a finite dimensional $F$-algebra. Then $Der\mathfrak{U}$ contains the semisimple and nilpotent parts of all its elements.*

pf. If $\delta \in \mathrm{Der}\mathfrak{U}$, let $\sigma, \nu \in \mathrm{End}\mathfrak{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathfrak{U}_a = \{x \in \mathfrak{U} | (\delta - a1)^k x = 0,\ \exists k\}$. Then $\mathfrak{U}$ is the direct sum of those $\mathfrak{U}_a$ for which $a$ is an eigenvalue of $\delta$ and $\sigma$ acts on $\mathfrak{U}_a$ as scalar multiplication by $a$. Then $\forall a, b \in F, \mathfrak{U}_a \mathfrak{U}_b \subset \mathfrak{U}_{a+b}$ since $(\delta - (a+b)1)^n (xy) = \sum_{i=0}^{n} ((\delta - a1)^{n-i} x) \cdot ((\delta - b1)^i y)$, $x, y \in \mathfrak{U}$. Then if $x \in \mathfrak{U}_a, y \in \mathfrak{U}_b$, $\sigma(xy) = (a+b)xy$, since $xy \in \mathfrak{U}_{a+b}$. On the other hand, $(\sigma x)y + x(\sigma y) = (a+b)xy$. By the directness of the sum $\mathfrak{U} = \coprod \mathfrak{U}_a$, $\sigma$ is a derivation. Since $\nu = \delta - \sigma$ and both $\delta, \sigma \in \mathrm{Der}\mathfrak{U}$, so is $\nu$.

**Lemma 1.2.3.** *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) | [xB] \subset A\}$. Suppose $x \in M$ satisfies $T(xy) = 0$, $\forall y \in M$. Then $x$ is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of $x$. Fix a basis $v_1, \ldots, v_m$ of $V$ relative to which $s$ has matrix $\mathrm{diag}(a_1, \ldots, a_m)$. Let $E$ be the vector subspace of $F$ over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \ldots, a_m$. Since $E$ has finite dimension over $\mathbb{Q}$, we have the dual space $E^*$ consisting of linear functions $f : E \to \mathbb{Q}$. Given $f$, let $y$ be the element of $\mathfrak{gl}(V)$ whose matrix relative to the given basis is $\mathrm{diag}(f(a_1), \ldots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$, $\begin{cases} \mathrm{ad}s(e_{ij}) = (a_i - a_j)e_{ij} \\ \mathrm{ad}y(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}$. Let $r(T) \in F[T]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$ for all pairs $i, j$. The existence follows from the Lagrange interpolation. Evidently $\mathrm{ad}y = r(\mathrm{ad}s)$. Then $\mathrm{ad}s$ is the semisimple part of $\mathrm{ad}x$ by the lemma 1.2.2., so it can be written as a polynomial in $\mathrm{ad}x$ without constant term by the proposition 0.2.1. Therefore $\mathrm{ad}y$ is also a polynomial in $\mathrm{ad}x$ without constant term. By the hypothesis, $\mathrm{ad}x$ maps $B$ into $A$ so $\mathrm{ad}y(B) \subset A$, i.e. $y \in M$. Then since $y = \mathrm{diag}(f(a_i))$ and only diagonal part of $x$ contributes to $T(xy)$, $T(xy) = 0$ and $\sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$-linear combination of elements of $E$, and applying $f$ gives $\sum f(a_i)^2 = 0$, but the numbers $f(a_i) \in \mathbb{Q}$ so all of them are forced to be 0. Thus $f$ must be identically 0, i.e. $E^*$ is zero, and therefore so is $E$.

**CLAIM : If $x, y, z$ are endomorphisms of a finite dimensional vector spaces, $T([xy]z) = T(x[yz])$.**
pf. $T([xy]z) = T(xyz - yxz) = T(xyz - xzy) = T(x[yz])$ since $T(y(xz)) = T((xz)y)$.

**Theorem 1.2.2** (Cartan's Criterion). *Let $L$ be a subalgebra of $\mathfrak{gl}(V)$, $V$ finite dimensional. Suppose $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. Then $L$ is solvable.*

pf. Let $A = [LL]$, $B = L$ so $M = \{x \in \mathfrak{gl}(V) | [xL] \subset [LL]\}$. Then $L \subset M$. By the hypothesis $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. If $[xy]$ is a typical generator of $[LL]$ and if $z \in M$, then by the previous claim, $T([xy]z) = T(x[yz]) = T([yz]x)$. Then by the definition of $M$, $[yz] \in [LL]$, so it is 0 by hypothesis. So by the previous lemma, $x \in [LL]$ are nilpotent so the $\mathrm{ad}_{[LL]}x$ is nilpotent. Thus $[LL]$ is nilpotent by Engel's theorem and $[LL]$ is solvable. Since $L/[LL]$ is abelian, the theorem follows by the previous proposition 1.1.3..

**Corollary 1.2.4.** *Let $L$ be a Lie algebra s.t. $T(adxady) = 0$, $\forall x \in [LL]$, $\forall y \in L$. Then $L$ is solvable.*

pf. By the previous theorem, the adjoint representation of $L$ gives $\mathrm{ad}L$ solvable. Since $\ker \mathrm{ad} = Z(L)$ is solvable given the centre being abelian, $L$ itself is solvable by the previous proposition 1.1.3.

**Definition 1.2.1.** *Let $L$ be any Lie algebra. If $x, y \in L$, define $\kappa(x,y) = T(adxady)$.*

$\to \kappa$ is associative, in the sense that $\kappa([xy],z) = \kappa(x,[yz])$.

**Lemma 1.2.4.** *Let $I$ be an ideal of $L$. If $\kappa$ is the killing form of $L$ and $\kappa_I$ the killing form of $I$, then $\kappa_I = \kappa|_{I\times I}$.*

pf. If $x, y \in I$, $(\mathrm{ad}x)(\mathrm{ad}y)$ is an endomorphism of $L$, mapping $L$ into $I$, so its trace $\kappa(x,y)$ coincides with the trace $\kappa_I(x,y)$ of $(\mathrm{ad}x)(\mathrm{ad}y)|_I = (\mathrm{ad}_I x)(\mathrm{ad}_I y)$.

A symmetric bilinear form $\beta(x,y)$ is called **nondegenerate** if its radical $S$ is 0, where $S = \{x \in L | \beta(x,y) = 0,\ \forall y \in L\}$. Because the killing form is associative, $S$ is an ideal of $L$.

We can test nondegeneracy as follows; fix a basis $x_1,\dots,x_n$ of $L$. Then $\kappa$ is nondegenerate iff the $n \times n$ matrix whose $i,j$ entry is $\kappa(x_i,x_j)$ has nonzero determinant.

e.g. $L = \mathfrak{sl}(2,F)$ with the standard basis $(x,h,y)$ gives $\mathrm{ad}h = \mathrm{diag}(2,0,-2)$, $\mathrm{ad}x = \begin{bmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $\mathrm{ad}y = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ and therefore $\kappa = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 0 \end{bmatrix}$ with determinant -128, and $\kappa$ is nondegenerate, given $\mathrm{char}F \neq 2$.

A Lie algebra $L$ is called semisimple in case $\mathrm{Rad}L = 0 \Leftrightarrow L$ has no nonzero abelian ideals. Conversely, the radical includes such an ideal of $L$, the last nonzero term in the derived series of $\mathrm{Rad}L$.

**Theorem 1.2.3.** *Let $L$ be a Lie algebra. Then $L$ is semisimple iff its killing form is nondegenerate.*

pf.

- Suppose $\mathrm{Rad}L = 0$. Let $S$ be the radical of $\kappa$. Then $T(\mathrm{ad}x\mathrm{ad}y) = 0$, $\forall x \in S,\ y \in L$. By the Cartan's Criterion, $\mathrm{ad}_L S$ is solvable, hence $S$ is solvable. Since $S$ is an ideal of $L$, $S \subset \mathrm{Rad}L = 0,$, $\kappa$ is nondegenerate.
- Suppose $S = 0$ and for any abelian ideal $I$ of $L$, $x \in I$, $y \in L$. Then $\mathrm{ad}x\mathrm{ad}y$ maps $L \to L \to I$ and $(\mathrm{ad}x\mathrm{ad}y)^2$ maps $L$ into $[II] = 0$, i.e. $\mathrm{ad}x\mathrm{ad}y$ is nilpotent, hence $0 = T(\mathrm{ad}x\mathrm{ad}y) = \kappa(x,y)$, so $I \subset S = 0$. So $L$ is semisimple.

$\to S \subset \mathrm{Rad}L$, but not vice versa.

**Definition 1.2.2.** *A Lie algebra $L$ is said to be the direct sum of ideals $I_1,\dots,I_t$ provided $L = I_1 + \cdots + I_t$ and written as $L = I_1 \oplus \cdots \oplus I_t$.*

$\to [I_iI_j] \subset I_i \cap I_j = 0$, $\forall i \neq j$.

**Theorem 1.2.4.** *Let $L$ be semisimple. Then there exist ideals $L_1,\ldots,L_t$ of $L$ which are simple, s.t. $L = L_1 \oplus \cdots \oplus L_t$. Every simple ideal of $L$ coincides with one of the $L_u$. Moreover, the killing form of $L_i$ is the restriction of $\kappa$ to $L_i \times L_i$.*

pf.

- **For an arbitrary ideal $I$**, $L = I \oplus I^{\perp}$

  Let $I$ be an arbitrary ideal of $L$. Then $I^{\perp} = \{x \in L | \kappa(x,y) = 0,\ \forall y \in I\}$ is also an ideal by the associativity of $\kappa$. By Cartan's criterion, the ideal $I \cap I^{\perp}$ of $L$ is solvable and therefore $L = I \oplus I^{\perp}$ given $\dim I + \dim I^{\perp} = \dim L$.

- existence
  If $L$ has no nonzero proper ideal, then $L$ is simple and there is nothing to prove.
  Otherwise, let $L_1$ be a minimal nonzero ideal. Then by the stmt above, $L = L_1 \oplus L_1^{\perp}$. In particualr, any ideal of $L_1$ is also an ideal of $L$, so $L_1$ is semisimple, hence simple by minimality. For the same reason, $L_1^{\perp}$ is semisimple. Then by inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of $L$. So the decomposition of $L$ follows.

- uniqueness

  If $I$ is any simple ideal of $L$, $[IL]$ is also an ideal of $I$, nonzero, given $Z(L) = 0$. Thus $[IL] = I$. On the other hand, $[IL] = [IL_1] \oplus \cdots \oplus [IL_t]$, so all but one summand must be 0. Say $[IL_1] = I$. Then $I \subset L_1$ and $I = L_1$ given $L_1$ simple.

**Corollary 1.2.5.** *If $L$ is semisimple, then $L = [LL]$ and all ideals and homomorphic images of $L$ are semisimple. Moreover, each ideal of $L$ is a sum of certain simple ideals of $L$.*

**ad$L$ is an ideal in Der$L$.**
pf. $[\delta, \mathrm{ad}x] = \mathrm{ad}(\delta x),\ \forall x \in L,\ \delta \in \mathrm{Der}L$.

**Theorem 1.2.5.** *If $L$ is semisimple, then $adL = DerL$, i.e. every derivation of $L$ is inner.*

pf. Since $L$ is semisimple, $Z(L) = 0$. Therefore $L \to \mathrm{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \mathrm{ad}L$ itself has nondegenerate killing form by the previous theorem 1.2.3. If $D = \mathrm{Der}L$, $[DM] \subset M$ by the previous stmt. Thus $\kappa_M$ is the restriction to $M \times M$ of the killing form $\kappa_D$ of $D$. In particular, if $I = M^{\perp}$ is the subspace of $D$ orthogonal to $M$ under $\kappa_D$, then the nondegeneracy of $\kappa_M$ forces $I \cap M = 0$. Since both $I, M$ are ideals of $D$, $[IM] = 0$. If $\delta \in I$, $\mathrm{ad}(\delta x) = 0,\ \forall x \in L$ by

the previous stmt so $\delta x = 0$, given ad is one to one and $\delta = 0$. Thus $I = 0$, Der$L$ = $M$ = ad$L$.
$\rightarrow$ since Der$L$ coincides with ad$L$, each $x \in L$ determines unique elements $s, n \in L$ s.t. ad$x$ = ad$s$ + ad$n$ the Jordan decomposition of ad$x$ in End$L$, given $L \rightarrow$ ad$L$ being one to one. This means $x = s + n$ with $[sn] = 0$, $\begin{cases} s \text{ ad-semisimple} \\ n \text{ ad-nilpotent} \end{cases}$. Write $s = s_x, n = n_x$ and call these the semisimple and nilpotent parts of $x$.

**Definition 1.2.3.** *Let $L$ be a Lie algebra. A vector space $V$, endowed with an operation* $\begin{matrix} L & \times & V \rightarrow V \\ (x,v) & \mapsto & xv \end{matrix}$ *is called an $L$-module if the following conditions are satisfied;* $\begin{cases} (ax+by)v = a(xv)+b(yv) \\ x(av+bw) = a(xv)+b(xw) \\ [xy]v = xyv - yxv \end{cases}$, $\begin{matrix} x,y \in L \\ v,w \in V \\ a,b \in F \end{matrix}$.

e.g. for $\phi : L \rightarrow \mathfrak{gl}(V)$, a representation of $L$, $V$ may be viewed as an $L$-module via the action $xv = \phi(x)(v)$. Conversely, given an $L$-module $V$, this equation defines a representation $\phi : L \rightarrow \mathfrak{gl}(V)$.

**Definition 1.2.4.** *A homomorphism of $L$-modules is a linear map $\phi : V \rightarrow W$ s.t. $\phi(xv) = x\phi(v)$.*

$\rightarrow \ker\phi$ is an $L$-submodule of $V$.

An $L$-module $V$ is called **irreducible** if it has precisely two $L$-submodules ( itself and 0 ). $V$ is called **completely reducible** if $V$ is a direct sum of irreducible $L$-submodules, or equivalently if each $L$-submodule $W$ of $V$ has a complement $W'$.

Given a representation $\phi : L \rightarrow \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in End$V$ leaves invariant precisely the same subspaces as $L$. Thus all the usual results for modules over associative rings hold for $L$.
**Schur's Lemma: Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be irreducible. The the only endomorphisms of $V$ commuting with all $\phi(x)$, $\forall x \in L$ are scalars.**

- Let $V$ be an $L$-module. The the dual vector space $V^*$ becomes an $L$-module called **contragredient** if we define $\forall f \in V^*$, $\forall v \in V$, $\forall x \in L$, $(xf)(v) = -f(xv)$.
  pf.
  $$\begin{aligned} ([xy]f)(v) &= -f([xy]v) && \text{by definition of operation} \\ &= -f(xyv - yxv) && \text{by the definition of bracket} \\ &= -f(xyv) + f(yxv) && \text{by the linearity} \\ &= (xf)(yv) - (yf)(xv) && = -(yxf)(v) + (xyf)(v) \\ &= ((xy - yx)f)(v) \end{aligned}$$
  so it is a module.

- If $V, W$ are $L$-modules, let $V \otimes W$ be the tensor product over $F$ of the underlying vector spaces. Then $x(v \otimes w) = xv \otimes w + v \otimes xw$, hence it is a module.
  pf. $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) =$

$(xyv\otimes w+v\otimes xyw)-(yxv\otimes w+v\otimes yxw)=(xy-yx)(v\otimes w)$, hence it is a module.

e.g. Given a vector space $V$ over $F$, $\begin{array}{ccc} V^*\otimes V & \to & \mathrm{End}V \\ f\otimes v & \mapsto & (w\mapsto f(w)v) \end{array}$. If in addition $V$ is an $L$-module, then $V^*\otimes V$ is an $L$-module in the way described above. Thus End$V$ can also be viewed as an $L$-module, s.t. $(xf)(v)=xf(v)-f(xv)$, $\forall x\in L\ f\in \mathrm{End}V,\ v\in V$.

Let $L$ be semisimple and $\phi: L\to \mathfrak{gl}(V)$ be faithful representation of $L$. Define a symmetric bilinear form $\beta(x,y)=T(\phi(x)\phi(y))$ on $L$. Then $\beta$ is associative, so its radical $S$ is an ideal of $L$. Moreover $\beta$ is nondegenerate ( by the Cartan's criterion, $\phi(S)\cong S$ is solvable, so $S=0$ ). Now let $L$ be semisimple, $\beta$ be any nondegenerate symmetric associative bilinear form on $L$. If $(x_1,\ldots,x_n)$ is a basis of $L$, there is a uniquely determined dual basis $(y_1,\ldots,y_n)$ relative to $\beta$ s.t. $\beta(x_i,y_j)=\delta_{ij}$. If $x\in L$,

$$\left\{\begin{array}{l} [xx_i]=\sum_j a_{ij}x_j \\ [xy_i]=\sum_j b_{ij}y_j \end{array}\right. \text{. Then } \begin{array}{rll} a_{ik}=\sum_j a_{ij}\beta(x_j,y_k)=\beta([xx_i],y_k) & & \text{by the definition } [xx_i]=\sum_j a_{ij}x_j \\ =\beta(-[x_ix],y_k) & & \text{by the definition} \\ =\beta(x_i,-[xy_k]) & & \text{by the associativity of } \beta \\ =-\sum_j b_{kj}\beta(x_i,y_j) & & [xy_i]=\sum_j b_{ij}y_j \\ =-b_{ki} & & \end{array}.$$

If $\phi: L\to\mathfrak{gl}(V)$ is any representation of $L$, write $c_\phi(\beta)=\sum_i\phi(x_i)\phi(y_i)\in \mathrm{End}V$ where $x_i,y_i$ run over dual bases relative to $\beta$. Then

$$\begin{array}{rll} [\phi(x),c_\phi(\beta)]=\sum_i[\phi(x),\phi(x_i)]\phi(y_i)+\sum_i\phi(x_i)[\phi(x),\phi(y_i)] & & \text{by the definition} \\ =\sum_{ij}a_{ij}\phi(x_j)\phi(y_i)+\sum_{ij}b_{ij}\phi(x_i)\phi(y_j) & & \text{by the identity } [x,yz]=[xy]z+y[xz] \\ =0 & & \text{by the identity } a_{ik}=-b_{ki} \end{array},$$

i.e. $c_\phi(\beta)$ is an endomorphism of $V$ commuting with $\phi(L)$.

**Definition 1.2.5.** *Let $\phi: L\to\mathfrak{gl}(V)$ be a faithful representation with trace form $\beta(x,y)=T(\phi(x)\phi(y))$. Having fixed a basis $(x_1,\ldots,x_n)$ of L, we call $c_\phi=c_\phi(\beta)$ the Casimir element of $\phi$.*

$$\to\left\{\begin{array}{c} \text{its trace is } \sum_i T(\phi(x_i)\phi(y_i))=\sum_i\beta(x_i,y_i)=\dim L \\ \text{if } \phi \text{ is irreducible, by the Schur's lemma } c_\phi \text{ is scalar} \end{array}\right..$$

e.g. $L=\mathfrak{sl}(2,F)$ with the standard basis $(x,h,y)$, $V=F^2$, $\phi: L\to\mathfrak{gl}(V)$ the identity map. Then the dual basis relative to the trace form is $(y,h/2,x)$, so $c_\phi=xy+h^2/2+yx=\begin{bmatrix}3/2 & 0\\ 0 & 3/2\end{bmatrix}$ where $3/2=\dim L/\dim V$.

When $\phi$ is not faithful, since $\ker\phi$ is an ideal of $L$, a sum of certain simple ideals. Let $L'$ denote the sum of the remaining simple ideals by the theorem 1.2.4. Then the restriction of $\phi$ to $L'$ is a faithful representation of $L'$ and we can use the previous argument of faithful representation.

**Lemma 1.2.5.** *Let $\phi: L\to\mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L. Then $\phi(L)\subset\mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L-module.*

pf. Since $L = [LL]$, given $L$ semisimple, and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$, the lemma follows.

**Theorem 1.2.6** (Weyl). *Let $\phi : L \to \mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then $\phi$ is completely reducible.*

pf.

- $V$ has an $L$-submodule $W$ of codimension 1.

  Since by the previous lemma $L$ acts trivially on $V/W$, we may denote this module $F$ and we get a s.e.s. $0 \to W \to V \to F \to 0$.

  - We can reduce the case into the one where $W$ is an irreducible $L$-module.

    Let $W'$ be a proper nonzero submodule of $W'$. This yields a s.e.s $0 \to W/W' \to V/W' \to F \to 0$. By the inductive hypothesis, this splits, i.e. there exists a 1-dimensional $L$-submodule $\tilde{W}/W'$ of $V/W'$ complementary to $W/W'$. So we have another s.e.s. $0 \to W' \to \tilde{W} \to F \to 0$ with $\dim W' < \dim W$. Thus by inductive hypothesis, $\tilde{W} = W' \oplus X$, for some submodule $X$ complementary to $W'$. But $V/W' = W/W' \oplus \tilde{W}/W'$, and therefore $V = W \oplus X$ given $\begin{cases} \dim W + \dim X = \dim V \\ W \cap X = 0 \end{cases}$.

  So assume $W$ is irreducible. Let $c = c_\phi$ be the Casimir element of $\phi$. Since $c$ commutes with $\phi(L)$, $c$ is actually an $L$-module endomorphism of $V$ and in particular $c(W) \subset W$ and $\ker c$ is an $L$-submodule of $V$. Because $L$ acts trivially on $V/W$, $c$ must do likewise. So $c$ has trace 0 on $V/W$. On the other hand, $c$ acts as a scalar on the irreducible $L$-submodule $W$ by Schur's lemma, which cannot be 0 because that would force $T_V(c) = 0$, a contradiction. Thus $\ker c$ is a 1-dimensional $L$-submodule of $V$ ( given $c|_W$ invertible, $\ker c \cap W = 0$ and $\dim \ker c \geq \dim V - \dim W = 1$ and $\ker c \cap W = 0$, $\dim W = \dim V - 1$ forces $\dim \ker c = 1$ ) which intersects $W$ trivially. This is the desired complement to $W$.

- General case

  Let $W$ be a nonzero submodule of $V$. Then $\exists$ a s.e.s. $0 \to W \to V \to V/W \to 0$. Let $\hom(V, W)$ be the space of linear maps $V \to W$, viewed as an $L$-module. Let $\mathfrak{V}$ be the subspace of $\hom(V, W)$ consisting of those maps whose restriction to $W$ is a scalar multiplication. Then $\forall f \in \mathfrak{V}$, $f|_W = a1_W$. So $\forall x \in L$, $\forall w \in W$, $(xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \to xf|_W = 0$, i.e. $\mathfrak{V}$ is an $L$-submodule. Let $\mathfrak{W}$ be the subspace of $\mathfrak{V}$ consisting of $f$ whose restriction to $W$ is zero. Then $\mathfrak{W}$ is also an $L$-submodule and $L$ maps $\mathfrak{V}$ into $\mathfrak{W}$. Moreover $\mathfrak{V}/\mathfrak{W}$ has dimension 1, because each $f \in \mathfrak{V}$ is determined by the scalar $f|_W \mod \mathfrak{W}$. Then we have a s.e.s. $0 \to \mathfrak{W} \to \mathfrak{V} \to F \to 0$ and we can use the previous case to get one dimensional submodule complementary to $\mathfrak{W}$. Let $f : V \to W$ span one dimensional complement to $\mathfrak{W}$ in $\mathfrak{V}$. Then by the previous lemma $L$ acts trivially on that 1-dimensional submodule. So $xf = 0$, $\forall x \in L$ so $f$ is an $L$-homomorphism. After multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. Since $f$ maps $V$ into $W$ and acts as $1_W$ on $W$, $V = W \oplus \ker f$.

**Theorem 1.2.7.** *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra. Then $L$ contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decomposition in $L$ coincide.*

pf. Let $x \in L$ be arbitrary with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\mathrm{ad}x(L) \subset L$, $\begin{cases} \mathrm{ad}x_s(L) \subset L \\ \mathrm{ad}x_n(L) \subset L \end{cases}$ by the proposition 1.2.1., where $\mathrm{ad} = \mathrm{ad}_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including $L$ as an ideal. Define $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W,\ T(y|_W) = 0\}$. For example $L_V = \mathfrak{sl}(V)$. Since $L = [LL]$, $x|_W \in \mathfrak{sl}(W)$ so $T(x|_W) = 0$ and therefore $L$ lies in all such $L_W$. Set $L' = N \cap (\cap_W L_W)$. Clearly, $L'$ is a subalgebra of $N$ including $L$ as an ideal and moreover if $x \in L$, $x_s, x_n \in L_W$ and therefore in $L'$. By the Weyl's theorem, given $L$ an ideal, $L' = L + M$ for some $L$-submodule $M$ where the sum is direct. But $[LL'] \subset L$, given $L' \subset N$, so the action of $L$ on $M$ is trivial. Let $W$ be any irreducible $L$-submodule of $V$. If $y \in M$, $[Ly] = 0$ so by the Schur's lemma, $y$ acts on $W$ as a scalar. On the other hand $T(y|_W) = 0$, given $y \in L_W$. Thus $y$ acts on $W$ as zero. $V$ can be written as a direct sum of irreducible $L$-submodules by the Weyl's theorem, so $y = 0$. This implies $M = 0$, $L = L'$ and the theorem follows.

**Corollary 1.2.6.** *Let $L$ be a semisimple Lie algebra $\phi : L \to \mathfrak{gl}(V)$ a representation of $L$. If $x = s+n$ is the abstract Jordan decomposition of $x \in L$, then $\phi(x) = \phi(s)+\phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. $\mathrm{ad}s$ is semisimple on $L$, so $L$ is spanned by $\mathrm{ad}x$-eigenvectors. By applying $\phi$, the algebra $\phi(L)$ is spanned by the eigenvectors of $\mathrm{ad}_{\phi(L)}\phi(s)$, since $L$ has this property relative to $\mathrm{ad}s$ and therefore $\mathrm{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\mathrm{ad}_{\phi(L)}\phi(n)$ is nilpotent and it commutes with $\mathrm{ad}_{\phi(L)}\phi(s)$. Accordingly $\phi(x) = \phi(s) + \phi(n)$ is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

## 1.3 Roots System

### 1.3.1 Root Space Decomposition

If $L$ consisted entirely nilpotent elements, then $L$ would be nilpotent by Engel's Theorem. Otherwise, we can find $x \in L$ s.t. whose semisimple part $x_s$ in the abstract Jordan decomposition is nonzero, i.e. $L$ possesses nonzero subalgebras consisting of semisimple elements. Such a subalgebra is called **toral**.

**Lemma 1.3.1.** *A toral subalgebra of L is abelian.*

pf. Let $T$ be toral. Since ad$x$ is semisimple and $F$ is algebraically closed, ad$x$ is diagonalisable. Suppose $[xy] = ax,\ a \neq 0$, i.e. $\mathrm{ad}_T x$ has nonzero eigenvalue. Then $\mathrm{ad}_T y(x) = -ay$ is itself an eigenvector of $\mathrm{ad}_T y$ of eigenvalue 0. On the other hand, we can write $x$ as a linear combination of eigenvectors of $\mathrm{ad}_T y$, given $y$ being semisimple. After applying $\mathrm{ad}_T y$ to $x$, all that is left is a combination of eigenvectors which belong to nonzero eigenvalues if any, a contradiction.

Fix a maximal toral subalgebra $H$ of $L$. For example, if $L = \mathfrak{sl}(n,F)$ has $H$, the set of diagonal matrices of trace 0. Since $H$ is abelian by the previous lemma, $\mathrm{ad}_L H$ is a commuting family of semisimple endomorphisms of $L$. Then $\mathrm{ad}_L H$ is simultaneously diagonalisable, i.e. $L$ is the direct sum of the subspaces $L_\alpha = \{x \in L | [hx] = \alpha(h)x,\ \forall h \in H\}$, where $\alpha$ ranges over $H^*$. The set of all nonzero $\alpha \in H^*$ for which $L_\alpha \neq 0$ is denoted by $\Phi$, whose elements are called the **roots** of $L$ relative to $H$.

**Definition 1.3.1.** *A root space decomposition or Cartan decomposition is defined as*

$$L = C_L(H) \oplus \coprod_{\alpha \in \Phi} L_\alpha$$

**Proposition 1.3.1.** $\forall \alpha, \beta \in H^*,\ [L_\alpha L_\beta] \subset L_{\alpha+\beta}$. *If* $x \in L_\alpha,\ \alpha \neq 0$, *then adx is nilpotent. If* $\alpha, \beta \in H^*,\ \alpha + \beta \neq 0$, *then* $L_\alpha$ *is orthogonal to* $L_\beta$ *relative to the killing form* $\kappa$ *of L.*

pf. $\forall \begin{cases} x \in L_\alpha \\ y \in L_\beta \end{cases}, h \in H$ implies $\mathrm{ad}h([xy]) = [[hx]y] + [x[hy]] = \alpha(h)[xy] + \beta(h)[xy] = (\alpha+\beta)(h)[xy]$ by the Jacobi identity. So the first stmt follows. Then the second stmt is the immediate consequence of the first stmt. Find $h \in H$ for which $(\alpha+\beta)(h) \neq 0$.

Then if $x \in L_\alpha, y \in L_\beta$, $\begin{cases} \kappa([hx], y) = -\kappa([xh], y) = -\kappa(x, [hy]) \text{ by the associativity of } \kappa \\ \alpha(h)\kappa(x,y) = -\beta(h)\kappa(x,y) \\ (\alpha+\beta)(h)\kappa(x,y) = 0 \end{cases}$,

which implies $\kappa(x,y) = 0$.

**Corollary 1.3.1.** *The restriction of the killing form to $L_0 = C_L(H)$ is nondegenerate.*

pf. By the theorem 1.2.3., $\kappa$ is nondegenerate. On the other hand, $L_0$ is orthogonal to all $L_\alpha$, $\alpha \in \Phi$ by the previous proposition. If $z \in L_0$ is orthogonal to $L_0$ as well, $\kappa(z,L) = 0 \to z = 0$.

**Lemma 1.3.2.** *If $x, y$ are commuting endomorphisms of a finite dimensional vector space, with $y$ nilpotent, then $xy$ is nilpotent; in particular, $T(xy) = 0$.*

**Proposition 1.3.2.** *Let $H$ be a maximal toral subalgebra of $L$. Then $H = C_L(H)$.*

pf. Write $C = C_L(H)$.

- **CLAIM : $C$ contains the semisimple and nilpotent parts of its elements**

  pf. $x \in C_L(H)$ so ad$x$ maps the subspace $H$ of $L$ into the subspace 0. Then by the previous proposition 1.2.1., $(\mathrm{ad}x)_s, (\mathrm{ad}x)_n$ have the same property. But $\begin{cases} (\mathrm{ad}x)_s = \mathrm{ad}x_s \\ (\mathrm{ad}x)_n = \mathrm{ad}x_n \end{cases}$, so the claim follows.

- **CLAIM : All semisimple elements of $C$ lie in $H$.**

  pf. If $x$ is semisimple and centralises $H$, $H + Fx$ is toral. The sum of commuting semisimple elements is again semisimple and therefore by the maximality of $H$, $x \in H$.

- **CLAIM : The restriction of $\kappa$ to $H$ is nondegerate.**

  pf. Let $\kappa(h,H) = 0$, $h \in H$. If $x \in C$ is nilpotent, $\begin{cases} [xH] = 0 \\ \mathrm{ad}x \text{ is nilpotent} \end{cases}$ imply $T(\mathrm{ad}x\mathrm{ad}y) = 0$, $\forall y \in H$, or $\kappa(x,H) = 0$. Then by the previous claims, $\kappa(h,C) = 0$, and therefore $h = 0$ and the claim follows.

- **$C$ is nilpotent.**

  pf. If $x \in C$ is semisimple, $x \in H$ by the previous claim2 and $\mathrm{ad}_C x = 0$ is certainly nilpotent. On the other hand, if $x \in C$ is nilpotent, then $\mathrm{ad}_C x$ is a fortiori nilpotent. Now let $x \in C$ be arbitrary, $x = x_s + x_n$ and $x_s, x_n \in C$ by the previous claim1. Since $\mathrm{ad}_C x$ is the sum of commuting nilpotents and is therefore itself nilpotent. Therefore by the Engel's theorem, $C$ is nilpotent.

- Since $\kappa$ is associative and $[HC] = 0$, $\kappa(H,[CC]) = 0$. If $C$ is not abelian, $[CC] \neq 0$. Since $C$ is nilpotent by the previous claim4, $Z(C) \cap [CC] \neq 0$. Let $0 \neq z \in Z(C) \cap [CC]$. Then by the previous claim2, $z$ cannot be semisimple. Its nilpotent part $n$ is therefore nonzero and lies in $C$ by the previous claim 1, and therefore also lies in $Z(C)$ by the proposition 1.2.1. Then by the previous lemma, $\kappa(n,C) = 0$, a contradiction to the previous corollary. Thus $C$ is abelian. If $C \neq H$, $C$ contains a nonzero nilpotent element $x$ by the previous claim 1,2. By the previous lemma, $\kappa(x,y) = T(\mathrm{ad}x\mathrm{ad}y) = 0$, $\forall y \in C$, a contradiction to the previous corollary, so $C = H$ and the proposition follows.

**Corollary 1.3.2.** *The restriction of $\kappa$ to $H$ is nondegenerate.*

$\rightarrow H \sim H^*$ in the sense that $\forall \phi \in H^*$, it corresponds to the unique element $t_\phi \in H$ satisfying $\phi(h) = \kappa(t_\phi, h)$, $\forall h \in H$.

**Proposition 1.3.3.** *1. $\Phi$ spans $H^*$*

2. *if $\alpha \in \Phi$, $-\alpha \in \Phi$*
3. *Let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Then $[xy] = \kappa(x,y)t_\alpha$*
4. *If $\alpha \in \Phi$, $[L_\alpha L_{-\alpha}]$ is one dimensional with basis $t_\alpha$*
5. *$\alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0$, $\forall \alpha \in \Phi$*
6. *If $\alpha \in \Phi$ and $x_\alpha$ is any nonzero element of $L_\alpha$, then there exists $y_\alpha \in L_{-\alpha}$ s.t. $x_\alpha, y_\alpha, h_\alpha = [x_\alpha y_\alpha]$ span a three dimensional simple subalgera of $L$ isomorphic to $\mathfrak{sl}(2,F)$ via $x_\alpha \mapsto \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y_\alpha \mapsto \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h_\alpha \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$*
7. *$h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$, $h_\alpha = -h_{-\alpha}$*

pf.

- If $\Phi$ fails to span $H^*$, there exists nonzero $h \in H$ s.t. $\alpha(h) = 0$, $\forall \alpha \in \Phi$ by the duality. Then $[h, L_\alpha] = 0$, $\forall \alpha \in \Phi$. Since $[hH] = 0$, $[hL] = 0$ or $h \in Z(L) = 0$, a contradiction. So the stmt follows.

- Let $\alpha \in \Phi$. If $-\alpha \notin \Phi$, $L_{-\alpha} = 0$, $\kappa(L_\alpha, L_\beta) = 0$, $\forall \beta \in H^*$ by the proposition 1.3.1. Thus $\kappa(L_\alpha, L) = 0$, a contradiction to the nondegeneracy of $\kappa$.

- Let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Let $h \in H$ be arbitrary. Then $\kappa(h,[xy]) = \kappa([hx], y) = \alpha(h)\kappa(x,y) = \kappa(t_\alpha, h)\kappa(x,y) = \kappa(\kappa(x,y)t_\alpha, h) = \kappa(h, \kappa(x,y)t_\alpha)$, i.e. $H$ is orthogonal to $[xy] - \kappa(x,y)t_\alpha$ and therefore $[xy] = \kappa(x,y)t_\alpha$ by the previous corollary.

- By 3, $t_\alpha$ spans $[L_\alpha L_{-\alpha}]$, provided $[L_\alpha L_{-\alpha}] \neq 0$. Let $0 \neq x \in L_\alpha$. If $\kappa(x, L_{-\alpha}) = 0$, then $\kappa(x, L) = 0$, a contradiction to the nondegeneracy of $\kappa$. Therefore $\exists 0 \neq y \in L_{-\alpha}$ for which $\kappa(x,y) \neq 0$. Then by 3, $[xy] \neq 0$.

- Suppose $\alpha(t_\alpha) = 0$ so that $[t_\alpha x] = 0 = [t_\alpha y]$, $\forall x \in L_\alpha,\ y \in L_{-\alpha}$. By 4, we can find such $x, y$ satisfying $\kappa(x,y) \neq 0$. Modifying if necessary by a scalar, we may assume $\kappa(x,y) = 1$. Then $[xy] = t_\alpha$ by 3. Thus the subspace $S$ of $L$ spanned by $x, y, t_\alpha$ is a three dimensional solvable algebra, $S \cong \mathrm{ad}_L S \subset \mathfrak{gl}(L)$. In particular, $\mathrm{ad}_L s$ is nilpotent $\forall s \in [SS]$, so $\mathrm{ad}_L t_\alpha$ is both semisimple and nilpotent, i.e. $\mathrm{ad}_L t_\alpha = 0$. Thus $t_\alpha \in Z(L)$, a contradiction to the choice of $t_\alpha$.

- Given $0 \neq x_\alpha \in L_\alpha$, find $y \in L_{-\alpha}$ s.t. $\kappa(x_\alpha, y_\alpha) = \frac{2}{\kappa(t_\alpha, t_\alpha)}$, which is guaranteed by 5 and the fact that $\kappa(x_\alpha, L_{-\alpha}) \neq 0$. Set $h_\alpha = 2t_\alpha / \kappa(t_\alpha, t_\alpha)$. Then $[x_\alpha y_\alpha] = h_\alpha$ by 3. Moreover $[h_\alpha x_\alpha] = \frac{2}{\alpha(t_\alpha)}[t_\alpha x_\alpha] = \frac{2\alpha(t_\alpha)}{\alpha(t_\alpha)} x_\alpha = 2x_\alpha$ and similarly, $[h_\alpha y_\alpha] = -2y_\alpha$. So $x_\alpha, y_\alpha, h_\alpha$ span a three dimensional subalgebra of $L$ with the same multiplication table as $\mathfrak{sl}(2,F)$.

- $t_\alpha$ is defined by $\kappa(t_\alpha, h) = \alpha(h)$, $\forall h \in H$. Then $t_\alpha = -t_{-\alpha}$. Also by the definition of $h_\alpha$, $h_{-\alpha} = \frac{2t_{-\alpha}}{\kappa(t_{-\alpha}, t_{-\alpha})} = \frac{-2t_\alpha}{\kappa(t_\alpha, t_\alpha)} = -h_\alpha$.

**Proposition 1.3.4.** *1. $\alpha \in \Phi$ implies $\dim L_\alpha = 1$. In particular, $S_\alpha = L_\alpha + L_{-\alpha} + H_\alpha$ and for given nonzero $x_\alpha \in L_\alpha$, there exists a unique $y_\alpha \in L_{-\alpha}$ satisfying $[x_\alpha y_\alpha] = h_\alpha$*

2. *If $\alpha \in \Phi$, the only scalar multiples of $\alpha$ which are roots are $\pm\alpha$.*

3. *If $\alpha, \beta \in \Phi$, then* $\begin{cases} \beta(h_\alpha) \in \mathbb{Z} \\ \beta - \beta(h_\alpha)\alpha \in \Phi \end{cases}$

4. *If $\alpha, \beta, \alpha+\beta \in \Phi$, $[L_\alpha L_\beta] = L_{\alpha+\beta}$*

5. *Let $\alpha, \beta \in \Phi$, $\beta \neq \pm\alpha$. Let $r, q$ be the largest integers for which $\beta - r\alpha, \beta + q\alpha$ are roots respectively. Then $\forall \beta + i\alpha \in \Phi$, $-r \leq i \leq q$ and $\beta(h_\alpha) = r - q$.*

pf.

- 1,2

  For each pair of roots $\alpha, -\alpha$, let $S_\alpha \cong \mathfrak{Sl}(2,F)$ be a subalgebra of $L$ constructed in the previous proposition. By the Weyl's theorem and the theorem 1.2., we have a complete description of all finite dimensional $S_\alpha$-modules, in particular, we can describe $\mathrm{ad}_L S_\alpha$. Fix $\alpha \in \Phi$. Consider the subspace $M$ of $L$ spanned by $H$ along with all root spaces of the form $L_{c\alpha}$, $c \in F^*$. This is an $S_\alpha$-submodule of $L$, thanks to the proposition 1.3.1.The weights of $h_\alpha$ on

$M$ are integers 0 and $2c = c\alpha(h_\alpha)$. In particular, all $c$ occuring here must be integral multiples of 1/2. Now $S_\alpha$ acts trivially on $\ker\alpha$,a subspace of codimension 1 in $H$ complementary to $Fh_\alpha$, while $S_\alpha$ is an irreducible $S_\alpha$-submodule of $M$. Taken together, $\ker\alpha$ and $S_\alpha$ exhaust the occurences of the weight 0 for $h_\alpha$. Therefore, the only even weights occurring in $M$ are $0, \pm 2$. Thus $2\alpha$ is not a root, i.e. twice a root is never a root. But then $(1/2)\alpha$ cannot be a root either, so 1 cannot occur as a weight of $h_\alpha$ in $M$. Then $M = H + S_\alpha$. In particular, $\dim L_\alpha = 1$ so $S_\alpha$ is uniquely determined as the subalgebra of $L$ generated by $L_\alpha, L_{-\alpha}$, and the only multiples of a root $\alpha$ which are roots are $\pm\alpha$.

- 3,5

  Let $\beta \neq \pm\alpha$. Set $K = \sum_{i\in\mathbb{Z}} L_{\beta+i\alpha}$. Then each root space is one dimensional and no $\beta + i\alpha$ can equal 0, so $K$ is an $S_\alpha$-submodule of $L$, with one-dimensional weight spaces for the distinct integral weights $\beta(h_\alpha) + 2i$, $\beta + i\alpha \in \Phi$. Obviously, not both 0 and 1 can occur as weights of this form, so $K$ is irreducible. The highest weight must be $\beta(h_\alpha) + 2q$ if $q$ is the largest integer for which $\beta + q\alpha$ is a root. Similarly for the lowest integer $r$ and the roots $\beta - r\alpha$. The weights on $K$ form an arithmetic progression with difference 2, which implies the roots $\beta + i\alpha$ form a string, called the $\alpha$-string through $\beta$, $\beta - r\alpha, \ldots, \beta, \ldots, \beta + q\alpha$. Then $(\beta - r\alpha)(h_\alpha) = -(\beta + q\alpha)(h_\alpha)$ or $\beta(h_\alpha) = r - q$.

- 4

  If $\alpha, \beta, \alpha + \beta \in \Phi$, $\mathrm{ad}L_\alpha$ maps $L_\beta$ onto $L_{\alpha+\beta}$, i.e. $[L_\alpha L_\beta] = L_{\alpha+\beta}$.

# Chapter 2

# Representation of Lie Algebra